Tony RuiKang OuYang

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Current Position

Machine Learning Group @ University of Cambridge — *Cambridge, UK*
Research Assistant in Probabilistic Machine Learning — *Oct 2024 – present*
Supervisor: Prof. José Miguel Hernández-Lobato

- Generative Models, Neural Sampler, and Boltzmann Generator

Education

University of Cambridge — *Cambridge, UK*
MPhil in Machine Learning and Machine Intelligence — *Sep 2023 – Aug 2024*

- Graduate with **Distinction** (**Grade: 77%, Top 20%**)
- Course Advisor & Thesis Supervisor: Prof. José Miguel Hernández-Lobato
- Thesis: Energy-based Neural Sampler for Boltzmann Distribution (**Grade: 77.5%**)

University of Oxford — *Oxford, UK*
Visiting Student in Mathematics and Statistics — *Jan 2022 – Jul 2022*

- Supervisor: Prof. Gesine Reinert and Prof. Geoff Nicholls
- Fully funded by Harbin Institute of Technology, Shenzhen

Harbin Institute of Technology, shenzhen — *Shenzhen, CN*
BEng in Data Science — *Sep 2019 – Jun 2023*

- Graduate with **Distinction** (**Grade: 90%, Top 10%**)
- Thesis Supervisor: Yi Zhao
- Thesis: Implicit Local Node Embedding Synchronisation for Graph Autoencoders (**Grade: 90%**)

Experiences

The Alan Turing Institute — *London, UK*
Research Assistant — *Apr. 2022 - Jul. 2022*
Supervisor: Prof. Gesine Reinert

- Research about scalable Graph Neural Network and Federated Learning for data privacy

iUniSpace — *Shenzhen, CN*
Machine Learning Engineer Intern — *Jul 2021 - Sep 2021*

- Machine learning algorithms for Land Subsidence Detection based on satellite data

Publications

*Equal Contributions

[1] **R. K. OuYang***, B. Qiang*, J. M. Hernández-Lobato. BNEM:A Boltzmann Sampler Based on Bootstrapped Noised Energy Matching. In *Arxiv 2409.09787*.

[2] **R. K. OuYang**. Energy-based Neural Sampler for Boltzmann Distribution. *MPhil Thesis*.

[3] **R. K. OuYang**, A. Elliot, S. Limnios, M. Cucuringu, G. Reinert. L2G2G: A Scalable Local-to-Global Network Embedding with Graph Autoencoders. In *Complex Networks and their Applications 2023*.

Academic Services

Reviewing

- NeurIPS Foundation Models for Science workshop 2024
- ICLR 2025

Supervision and Teaching

Supervision for MEng Thesis — University of Cambridge
Sunaabh Trivedi — *Oct 2024 - present*

Selected Projects

The Family of Denoising Diffusion Probabilistic Models *Jan 2024 – Mar 2024*

- Replicate codes and exps of DDPM and its variants (DDIM&ImprovedDDPM) from scratch
- Research on extension for Improved DDPM
- Open source: GitHub Repo

Federated Curriculum Learning *Jan 2024 – Mar 2024*

- Research on applying Curriculum Learning scheme on Federated Learning: self-paced learning, teacher-transfer and deep mutual learning
- Open source: GitHub Repo

PPO in the Bayesian World *Nov 2023 – Jan 2024*

- Research on performing Bayesian Optimization for Reinforcement Learning (Proximal Policy Optimization, PPO) hyper-parameters tuning
- Research on using multi-fidelity simulations to speed up tuning
- Open source: GitHub Repo

Awards and Achievements

Distinction in MPhil Degree, University of Cambridge *2024*
*Ranked top **20%** in MPhil in Machine Learning and Machine Intelligence*

Travel and Research Grants, Wolfson College, University of Cambridge *2023*
Award for supporting my travelling for conference

Outstanding Graduates, Harbin Institute of Technology (Shenzhen) *2023*
*Ranked top **10**% in HITsz*

Distinction in BEng Degree, Harbin Institute of Technology (Shenzhen) *2023*
*Ranked top **10**% in BEng in Data Science*

International Exchange Program Scholarship, HITsz *2022*
*Scholarship for visiting in Oxford, valued at abount **200,000** CNY*

Others

Athletic

Player, Wolfson College (University of Cambridge) Football Varsity	*2023 - 2024*
Player, Harbin Institute of Technology (Shenzhen) Football Varsity	*2020 - 2023*
Captain, Harbin Institute of Technology (Shenzhen) Science School Football Team	*2021 - 2023*
2nd place of HITsz Football League Season 22-23	*2023*
2nd place of HITsz Football League Season 20-21	*2021*
2nd place of HITsz Futsal League Season 20-21	*2021*
3rd place of HITsz Football League Season 19-20	*2020*

Competitions

2nd Prize - Contemporary Undergraduate Mathematical Contest in Modeling	*2021*
1st Prize - National College Students Mathematical Competition	*2020*
2nd Prize - Contemporary Undergraduate Mathematical Contest in Modeling	*2020*

Services

Student Representative of Harbin Institute of Technology, shenzhen	*2021*

Languages

- Cantonese (1st Native)
- Chinese Mandarin (2nd Native)
- English (Professional working proficiency; IELT 7.0)
- French (Elementary proficiency)